

Leonardo Pelorosso

✉ pelorosso.leonardo@gmail.com | 🌐 Portfolio | 🐙 GitHub | Italian / EU citizen

M.Sc. candidate in High Performance Computing Engineering interested in parallel numerical methods, scientific machine learning, and performance-oriented software.

EDUCATION

Politecnico di Milano

2024 – Present

M.Sc. in High Performance Computing Engineering

Milan, Italy

- **Grade average:** 28.9/30 **ECTS earned:** 95/120
- **Expected graduation:** June 2026
- **Thesis availability:** 9 months from October 2026
- **Relevant coursework:** Parallel Computing, Computer Architecture, Advanced Methods for Scientific Computing, Numerical Analysis for Machine Learning, Deep Learning, Operations Research.

Politecnico di Milano

2021 – 2024

B.Sc. in Engineering Physics

Milan, Italy

- **Final grade:** 96/110

PROJECTS

Monte Carlo Integration & Optimization

Oct. 2025 – Jan. 2026

- Co-developed a high-performance C++ engine for N-dimensional Monte Carlo integration, including importance sampling, Metropolis–Hastings MCMC, runtime expression parsing with muParserX, and convex-polytope integration with Qhull.
- Parallelized Particle Swarm Optimization and Genetic Algorithms with OpenMP, achieving up to **10× speedup**; applied the solvers to drone center-of-mass balancing and a 15-turbine wind-farm layout problem.

AstraLog-HPC — Software Engineering

Mar. 2026 – June 2026

- Co-developed a Python/Polars pipeline for telemetry validation and stateful anomaly detection; jointly benchmarked processing and scalability on CINECA Galileo100.
- Contributed requirements analysis, OOP implementation, core pytest tests, GitHub Actions CI, and Docker/Singularity containerization for SLURM deployment.

Credit Default Prediction — Numerical ML

Sept. 2026

- Built a leakage-safe Python/JAX pipeline on 150,000 credit records, comparing regularized logistic regression, a Nyström-approximated kernel support vector machine, and a calibrated multilayer perceptron.
- Studied scaling, conditioning, kernel approximation, gradient verification, calibration, and cost-sensitive thresholds; evaluated models with stratified splits using average precision, ROC-AUC, Brier score, recall, and F1.

SKILLS

Languages: C/C++, Python, SQL, RISC-V Assembly

Scientific Computing & ML: NumPy, SciPy, JAX, PyTorch, scikit-learn, Polars

HPC: OpenMP, MPI, CUDA, Slurm

Engineering Tools: Git, Docker, Singularity, pytest, GitHub Actions

Databases: PostgreSQL, MongoDB, Neo4j (Cypher), Cassandra, Elasticsearch, Redis

Spoken Languages: Italian (native), English (C1, Cambridge CAE: 190), Spanish (B1)